

II. Missing Related Work: Succinct Light Clients, Bridges, & Accountability (2020–2026)

A. Direct Citations & Competitors to Our Work (CSSV)

- • **Practical Light Clients for Committee-Based Blockchains**
 - ◦ *Authors:* Frederik Armknecht, Ghassan Karame, Malcom Mohamed, Christiane Weis.
PDF
 - ◦ *Venue:* arXiv (October 2024).
PDF
 - ◦ *Relevance:* This paper directly analyzes and benchmarks against our protocol, referring to it as "CSSV". It proposes an alternative approach using transitive multi-signatures to verify block sequences in $O(1)$ time, assuming a static quorum of validators, claiming up to a $41,726\times$ reduction in latency compared to CSSV for short offline periods.
PDF+ 1
 - ◦ *Link:* [<https://arxiv.org/pdf/2410.03347>] (<https://arxiv.org/pdf/2410.03347>)
- • **Efficient Query Verification for Blockchain**

Superlight Clients Using SNARKs

- ◦ *Authors:* Stefano De Angelis, Ivan Visconti, Andrea Vitaletti, Marco Zecchini.
PDF
- ◦ *Venue:* Blockchain: Research and Applications (September/October 2025).
PDF
- ◦ *Relevance:* Highly relevant. The authors formalize a "Stateless Superlight Client" (SSLC) that outsources map-reduce application-specific queries to untrusted Oracle servers using recursive SNARKs (Plonky2/FRI) to prove computational correctness. The SSLC fetches public inputs (transaction roots/counts) from independent full nodes under an "existential honesty" assumption to cross-check the SNARK without downloading full blockchain data. It explicitly decouples consensus verification from query verification.
Nethermind+ 3
- ◦ *Link:* [<https://doi.org/10.1016/j.bcra.2025.100396>] (<https://doi.org/10.1016/j.bcra.2025.100396>)

- • **Trustless Bridges via Random Sampling Light Clients**
 - ◦ *Venue:* AFT 2025.
 - ◦ *Relevance:* Directly compares the gas costs of its on-chain random sampling verifier against SNARK-based approaches, explicitly citing our 2022 ePrint.
 - ◦ *Link:* <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.AFT.2025.31>
- • **Enhancing Blockchain Cross-Chain Interoperability: A Comprehensive Survey**
 - ◦ *Venue:* arXiv (May 2025).
 - ◦ *Relevance:* A systematization of knowledge that categorizes our paper within modern light-client interoperability paradigms.
 - ◦ *Link:* <https://arxiv.org/html/2505.04934v1>
- • **hinTS: Threshold Signatures with Silent Setup**
 - ◦ *Venue:* Hedera (2025).
 - ◦ *Relevance:* Cites our work in the context

of accountable signature systems and PLONK-based arguments.

- ◦ *Link:* [https://hedera.com/wp-content/uploads/2025/11/hinTS_Threshold_Signatures_with_Silent_Setup.pdf](https://hedera.com/wp-content/uploads/2025/11/hinTS_Threshold_Signatures_with_Silent_Setup.pdf)

B. Consensus Architecture & Overhead Analysis

- • **Lightweight Consensus in Blockchain: A Systematic Literature Review**
 - ◦ *Authors:* Namrata Marium Chacko, Narendra V G, Mamatha Balachandra, Manoj T.
 - ◦ *Venue:* ACM Computing Surveys, Vol. 58, Issue 3, Article No.: 78 (October 2025).
 - ◦ *Relevance:* Reviews 127 studies on lightweight blockchain consensus, proposing a novel taxonomy based on agreement methods and round propagation. This context is critical because lightweight consensus directly dictates the validator committee footprints and state validation overheads that our light client aims to compress.

Peeref

- ◦ *Link:* [<https://doi.org/10.1145/3768149>] (<https://doi.org/10.1145/3768149>)

C. Explicitly Requested by Reviewers

- • **zkBridge: Trustless Cross-chain Bridges Made Practical**
 - ◦ *Authors:* Tiancheng Xie, Jiaheng Zhang, Zerui Cheng, Fan Zhang, Yupeng Zhang, Yongzheng Jia, Dan Boneh, Dawn Song.
 - ◦ *Venue:* ACM CCS 2022.
 - ◦ *Link:* [<https://dl.acm.org/doi/10.1145/3548606.3560652>] (<https://dl.acm.org/doi/10.1145/3548606.3560652>)
- • **Nova: Recursive Zero-Knowledge Arguments from Folding Schemes**
 - ◦ *Authors:* Abhiram Kothapalli, Srinath Setty, Ioanna Tzialla.
 - ◦ *Venue:* CRYPTO 2022.
 - ◦ *Link:* [<https://eprint.iacr.org/2021/370>] (<https://eprint.iacr.org/2021/370>)
- • **Revisiting the Nova Proof System on a Cycle of Curves**
 - ◦ *Authors:* Wilson Nguyen, Dan Boneh,

Srinath Setty.

- ◦ *Venue:* IACR ePrint 2023.
 - ◦ *Link:* [<https://eprint.iacr.org/2023/969>] (<https://eprint.iacr.org/2023/969>)
- • **Traitor Tracing for Inner Product Functional Encryption with Compact Ciphertexts**
 - ◦ *Authors:* Various.
 - ◦ *Venue:* CRYPTO 2023. (*Represents the functional encryption and group signature traceability literature requested to address accountability*).
 - ◦ *Link:* [<https://eprint.iacr.org/2023/658>] (<https://eprint.iacr.org/2023/658>)

D. Commit-and-Prove SNARK Frameworks

- • **Lunar: A Toolbox for More Efficient Universal and Updatable zkSNARKs and Commit-and-Prove Extensions**
 - ◦ *Authors:* Matteo Campanelli, Antonio Faonio, Dario Fiore, Anaïs Querol, and Harkirat Singh.
 - ◦ *Venue:* PKC 2021.
 - ◦ *Link:* [<https://eprint.iacr.org/2021/100>] (<https://eprint.iacr.org/2021/100>)

eprint.iacr.org/2020/1069]
(<https://eprint.iacr.org/2020/1069>)

- • **Eclipse: Enhanced Compiling Method for Pedersen-Committed zkSNARK Engines**
 - ◦ *Authors:* Diego F. Aranha, Emil Madsen Bennedsen, Matteo Campanelli, Chaya Ganesh, Claudio Orlandi, and Akira Takahashi.
 - ◦ *Venue:* PKC 2022.
 - ◦ *Link:* [<https://eprint.iacr.org/2021/934>]
(<https://eprint.iacr.org/2021/934>)
- • **Doubly Aggregatable Signatures**
 - • *Authors:* Georg Fuchsbauer, Pranav Garimidi, Max Resnick, Joachim Neu, Ertem Nusret Tas, and Guru-Vamsi Policharla.
 - • *Venue:* Technical Report / Institutional Release (2026).
 - • *Relevance:* This work does **not** address cross-chain bridges or light clients. It is strictly scoped to an internal consensus-layer problem, specifically mitigating vote-withholding among validators by tracking a two-dimensional "who observed whom" attestation matrix. Furthermore, the protocol does not utilize standard BLS signatures; it introduces a custom two-layer,

pairing-based signature scheme grounded in the Random Modular Subset Sum (RMSS) assumption.

The sole intersection with our work is at the signature aggregation boundary. Both protocols confront the bottleneck of compressing massive signer subsets for commit-and-prove (CP) verification. However, their approaches diverge fundamentally: Fuchsbauer et al. invent a new algebraic signature primitive to allow a downstream, generic CP-SNARK to verify the matrix in poly-logarithmic time. In contrast, our Committee Key Scheme (CKS) retains standard cryptographic commitments for the validator vector and instead modifies the zero-knowledge architecture itself, utilizing a custom, monolithic hybrid-SNARK to achieve true $O(1)$ constant-time verification suitable for on-chain smart contracts.

- • *Link:* <https://github.com/guruvamsi-policharla/doubly-aggregatable-signatures>

E. State-of-the-Art Incremental Verification & Folding (2023–2025)

- • **SuperNova: Proving Universal Machine Executions without Universal Circuits**
 - ◦ *Authors:* Abhiram Kothapalli, Srinath Setty, Ioanna Tzialla.
 - ◦ *Venue:* IACR ePrint 2022 / Published 2023.
 - ◦ *Link:* [<https://eprint.iacr.org/2022/1758>]

(<https://eprint.iacr.org/2022/1758>)

- • **HyperNova: Recursive Arguments for Customizable Constraint Systems**
 - ◦ *Authors:* Abhiram Kothapalli, Srinath Setty.
 - ◦ *Venue:* CRYPTO 2024.
 - ◦ *Link:* [<https://eprint.iacr.org/2023/573>] (<https://eprint.iacr.org/2023/573>)
- • **ProtoGalaxy: Efficient ProtoStar-Style Folding of Multiple Instances**
 - ◦ *Authors:* Ariel Gabizon, Zachary J. Williamson.
 - ◦ *Venue:* IACR ePrint 2023.
 - ◦ *Link:* [<https://eprint.iacr.org/2023/1106>] (<https://eprint.iacr.org/2023/1106>)

F. Foundational Light Client & Systemization Papers

- • **Flyclient: Super-Light Clients for Cryptocurrencies**
 - ◦ *Authors:* Benedikt Bünz, Lucianna Kiffer, Loi Luu, and Mahdi Zamani.
 - ◦ *Venue:* IEEE S&P 2020.

- ◦ *Link:* [<https://eprint.iacr.org/2019/226>] (<https://eprint.iacr.org/2019/226>)
- • **SoK: A Review of Cross-Chain Bridge Hacks in 2023**
 - ◦ *Venue:* arXiv 2025.
 - ◦ *Link:* [<https://arxiv.org/html/2501.03423v1>] (<https://arxiv.org/html/2501.03423v1>)
- • **Formal Verification of a Fail-Safe Cross-Chain Bridge**
 - ◦ *Venue:* OASICS FMBC 2025.
 - ◦ *Link:* [<https://drops.dagstuhl.de/storage/01oasics/oasics-vol129-fmbc2025/OASICS.FMBC.2025.8/OASICS.FMBC.2025.8.pdf>] (<https://drops.dagstuhl.de/storage/01oasics/oasics-vol129-fmbc2025/OASICS.FMBC.2025.8/OASICS.FMBC.2025.8.pdf>)
- • **A Scalable Cross-Chain Data Asset Rights Confirmation Framework for Distributed Systems Based on Hybrid Post-Quantum Zero-Knowledge Proofs**
 - ◦ *Authors:* L. Zhang, Z. Yan.

- ◦ *Venue:* EAI Endorsed Transactions on Scalable Information Systems 2025.
- ◦ *Link:* <https://publications.eai.eu/index.php/sis/article/view/12050>